

# Exploratory Data Analysis (EDA)

E-Commerce Orders Dataset — Project 2

*DecodeLabs Data Analytics Internship | Batch 2026*

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# 1. Problem Statement

DecodeLabs collected twelve months of transactional order data from its e-commerce platform. Before any dashboarding or predictive modeling can be trusted, the raw data must be interrogated for structure, quality, and hidden patterns.

The core question this analysis answers: What do the order records tell us about product performance, customer payment behavior, order fulfillment, and revenue trends — and is the data clean and reliable enough to act on?

# 2. Methodology

Dataset: 1,200 order records across 14 fields (OrderID, Date, CustomerID, Product, Quantity, UnitPrice, ShippingAddress, PaymentMethod, OrderStatus, TrackingNumber, ItemsInCart, CouponCode, ReferralSource, TotalPrice), spanning January 2023 to June 2025.

## 2.1 Data Cleaning & Validation Steps

- Checked for missing values across all 14 columns.
- Checked for duplicate rows and duplicate Order IDs.
- Verified  $TotalPrice = Quantity \times UnitPrice$  for internal consistency.
- Computed descriptive statistics (mean, median, min, max, quartiles) for all numeric fields.
- Applied the IQR method ( $1.5 \times IQR$  rule) to flag statistical outliers in Quantity, UnitPrice, ItemsInCart, and TotalPrice.
- Grouped and aggregated revenue by Product, Order Status, Payment Method, Referral Source, and Coupon Code.
- Built a monthly time series of order volume and revenue to inspect trend behavior.

# 3. Key Findings

## 3.1 Data Quality Summary

| Metric                                           | Result                                                             |
|--------------------------------------------------|--------------------------------------------------------------------|
| Total records                                    | 1,200                                                              |
| Missing values                                   | Only in CouponCode (309 blanks = 25.8%, meaning no coupon applied) |
| Duplicate rows / Order IDs                       | 0 — dataset is fully unique                                        |
| Price consistency ( $Qty \times Price = Total$ ) | 100% match, no arithmetic errors                                   |
| Date range                                       | Jan 1, 2023 – Jun 30, 2025                                         |

Verdict: this dataset is clean. The only "missing" values are in CouponCode, which is expected — it simply means the order had no coupon applied, not a data error.

## 3.2 Descriptive Statistics (Five-Number Summary)

| Field            | Mean     | Median | Min   | Max      | Std Dev |
|------------------|----------|--------|-------|----------|---------|
| Quantity         | 2.95     | 3      | 1     | 5        | 1.41    |
| Unit Price (\$)  | 356.41   | 364.21 | 11.39 | 699.93   | 197.18  |
| Items in Cart    | 5.49     | 5      | 1     | 10       | 2.28    |
| Total Price (\$) | 1,053.97 | 823.62 | 11.39 | 3,456.40 | 819.86  |

Total Price is right-skewed — the mean (\$1,053.97) sits well above the median (\$823.62). This tells us a smaller number of large-basket, high-value orders are pulling the average upward. This is the same skew pattern seen in salary or house-price data: median is the more honest "typical order" figure, not the mean.

## 3.3 Outlier Detection (IQR Method)

Applying the  $1.5 \times IQR$  rule (values outside  $Q1 - 1.5 \times IQR$  to  $Q3 + 1.5 \times IQR$ ):

- Quantity, Unit Price, and Items in Cart: 0 outliers — these fields are well-behaved and bounded by design.
- Total Price: 8 orders exceed the upper bound of \$3,330. These are not data errors — they are large legitimate combinations of high quantity and high unit price (signal, not noise), consistent with bulk purchases.

### 3.4 Correlation Analysis

| Variable Pair              | Correlation (r) | Interpretation                                                          |
|----------------------------|-----------------|-------------------------------------------------------------------------|
| Unit Price ↔ Total Price   | 0.72            | Strong positive — price per item is the dominant driver of order value  |
| Quantity ↔ Total Price     | 0.62            | Moderate positive — more units ordered raises total spend               |
| Quantity ↔ Items in Cart   | 0.65            | Moderate positive — larger orders tend to have more distinct cart items |
| Unit Price ↔ Items in Cart | 0.001           | No relationship — cart size is independent of individual item price     |

As per the golden rule of EDA, these are correlations, not causal proof — but Unit Price is clearly the strongest lever on order value, more so than Quantity.

### 3.5 Product Performance

| Product | Orders | Total Revenue (\$) | Avg Order Value (\$) |
|---------|--------|--------------------|----------------------|
| Chair   | 178    | 195,620.11         | 1,098.99             |
| Printer | 181    | 195,612.61         | 1,080.73             |
| Laptop  | 173    | 192,126.56         | 1,110.56             |
| Tablet  | 179    | 186,568.95         | 1,042.28             |
| Monitor | 163    | 175,651.41         | 1,077.62             |
| Desk    | 170    | 167,459.93         | 985.06               |
| Phone   | 156    | 151,722.39         | 972.58               |

Chair and Printer lead in total revenue despite Laptop having the highest average order value (\$1,110.56) — Laptop's smaller order count (173) keeps its total revenue below Chair and Printer. Phone is the weakest performer on both order count and average value.

### 3.6 Order Status & Fulfillment

| Order Status | Count | % of Orders | Total Revenue (\$) |
|--------------|-------|-------------|--------------------|
| Cancelled    | 250   | 20.8%       | 276,396.21         |
| Returned     | 247   | 20.6%       | 243,277.70         |
| Pending      | 237   | 19.8%       | 256,328.15         |
| Shipped      | 235   | 19.6%       | 246,159.58         |
| Delivered    | 231   | 19.3%       | 242,600.32         |

This is the most significant business flag in the dataset: only 19.3% of orders are marked "Delivered." Cancelled (20.8%) and Returned (20.6%) orders combined represent over 41% of all order volume and over \$519,000 in revenue tied up in non-completed transactions — nearly double the delivered revenue's share of concern.

### 3.7 Payment Method & Marketing Channel

Payment methods are evenly distributed (Online 21.5%, Cash 20.5%, Credit Card 19.5%, Debit Card 19.3%, Gift Card 19.2%), with Credit Card orders carrying the highest average value (\$1,127.55). Referral sources are similarly balanced, with Instagram (259 orders, \$275,285 revenue) as the top-performing channel and Facebook showing the highest average order value (\$1,098.29) despite fewer orders.

### 3.8 Revenue Trend Over Time

Monthly order volume and revenue fluctuate without a strong seasonal spike — order counts per month range roughly between 27 and 53, with revenue moving in the same direction as volume (correlation is expected and confirmed). No single month dominates, suggesting steady, non-seasonal demand rather than holiday-driven spikes.

## 4. Recommendations

- Investigate the Cancelled/Returned segment (41.4% of orders): interview or survey these customers, cross-check with PaymentMethod and Product to identify if specific products or payment types drive higher cancellation/return rates.
- Use median, not mean, when reporting "typical order value" to leadership — the mean is inflated by a right-skewed tail of large orders.
- Treat the 8 high-value Total Price outliers as legitimate bulk orders worth a loyalty/VIP follow-up, not as errors to be removed.
- Double down on Unit Price–driven upsell strategies (e.g., premium product bundling) since Unit Price correlates most strongly with order value ( $r = 0.72$ ).
- Expand Instagram and Facebook referral investment — Instagram drives the most orders while Facebook drives the highest average order value.
- Since Laptop has the highest average order value but a lower order count, test targeted promotions to grow Laptop volume without diluting its higher basket size.

## 5. Conclusion

This dataset is structurally clean — no duplicates, no broken price logic, and only expected missingness in CouponCode. The real story is not in the data's quality but in its distribution: order value is right-skewed and price-driven, and a striking 41% of orders never reach "Delivered" status. That fulfillment gap is the single most actionable insight in this analysis and should be the starting point for any follow-up investigation or dashboard.